

Student Grade Analyzer

Scenario: You Are a School System Developer

Imagine yourself as a junior software developer working for a school that is digitizing its student grading system. Traditionally, the teachers have been calculating student results manually using notebooks and calculators. This process is slow, repetitive, and prone to human error.

Your task is to build a simple Python program that helps a teacher quickly analyze student performance for a class of 5 students. The school has given you a very basic requirement:

“We need a simple system where we enter marks for each student, and the system should automatically tell us who passed and who failed.”

A student is considered **PASS** if their mark is **50 or above**, otherwise they are considered **FAIL**.

Your Task

You will create a Python program that simulates this grading assistant. Your program should:

1. Collect Student Marks by asking the teacher (user) to enter marks for 5 students one by one and each input should represent one student's exam mark.

Example:

- Student 1 mark
- Student 2 mark
- Student 3 mark
- Student 4 mark
- Student 5 mark

2. Process Each Student's Result and for every student entered display the student number, the mark entered and determine whether the student has **Passed (≥ 50) or Failed (< 50)**

Example output:

- Student 1: 75 → Pass
- Student 2: 42 → Fail

3. Track Class Performance as you process the marks, your program should also be able to count how many students passed and keep track of all marks entered. This will help the teacher understand the overall class performance at a glance.

4. Display Final Summary after all 5 students have been processed, display a summary of the total number of students who passed

- Example:
- Total students who passed: 3

Expected Output Flow

Enter mark for student 1: 75

Student 1: 75 → Pass

Enter mark for student 2: 42

Student 2: 42 → Fail

Enter mark for student 3: 60

Student 3: 60 → Pass

Enter mark for student 4: 30

Student 4: 30 → Fail

Enter mark for student 5: 85

Student 5: 85 → Pass

Total students who passed: 3

Concepts You Must Use

This assignment is designed to help you practice the core Python fundamentals:

- `input()` → Getting user input
- `print()` → Displaying output
- `if / else` → Decision making
- `for` loop → Repeating tasks
- Variables → Storing values
- Integer conversion using `int()`

Hints (Think Like a Developer)

A good developer plans before writing code so before coding, think about:

- ❖ How will I repeat the process 5 times?
- ❖ How do I store the number of students who passed?
- ❖ Where do I apply the condition (pass/fail)?
- ❖ How do I make the output clear and readable?

Challenge yourself with these Bonus task (Optional)

Once you complete the main task, improve your program by adding the following features:

1. Calculate the average mark of the class.
2. Identify the highest mark entered.
3. Assign grades based on marks:
 - **A** → 80–100
 - **B** → 70–79
 - **C** → 60–69
 - **D** → 50–59
 - **E** → Below 50

Display the grade alongside each student's result.

Learning Outcome

By completing this challenge, you will be able to:

- Build a simple real-world Python application
- Use loops to automate repetitive tasks
- Apply conditional logic to decision-making problems
- Work with user input effectively
- Begin thinking like a software engineer solving real problems

Good luck and enjoy building your first grading system!